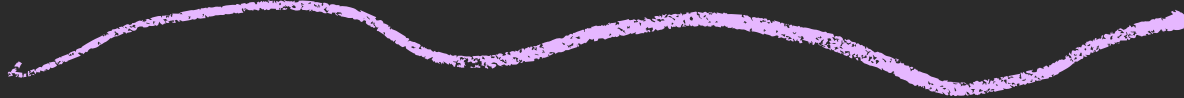




Image Processing with Open-CV



Agenda



Introduction

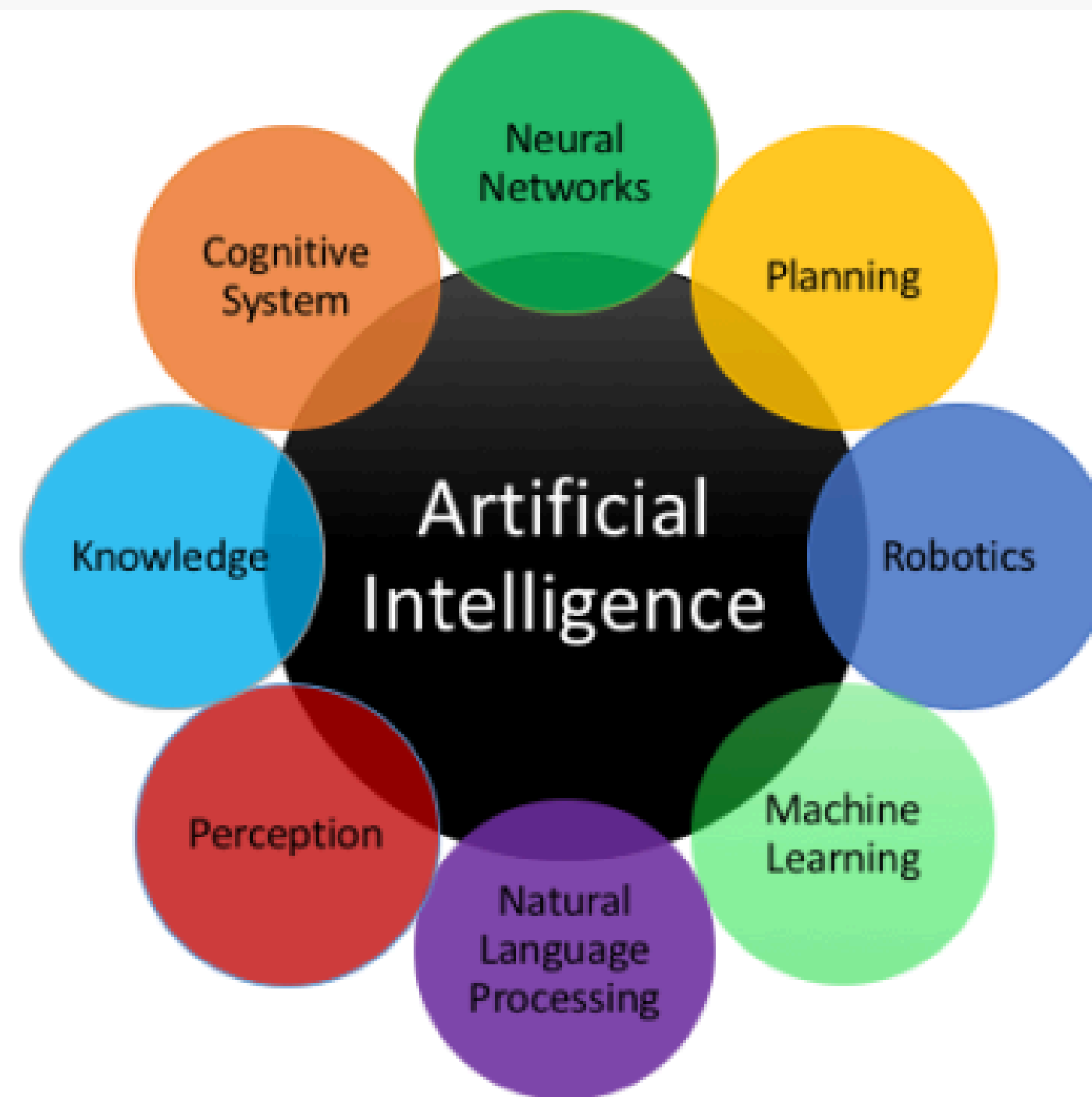
How does its works?

What is CV?

Application for CV

Why cv?

Installation for opencv



COMPUTER VISION

Introduction

Computer vision is the analysis of digital images by a computer for such applications as:

- **Industrial:** part localization and inspection, robotics
- **Medical:** disease classification, screening, planning
- **Military:** autonomous vehicles, tank recognition
- **Intelligence Gathering:** face recognition, video analysis
- **Security:** video analysis
- **Science:** classification, measurement
- **Document Processing:** text recognition, diagram conversion

What is ... ?



Computer Vision

Computer vision is a form of artificial intelligence that trains computers to receive information from digital and real-world images.

Computer vision trains machines to perform and automate tasks that typically require human intelligence.



Why CV?

computer vision becoming increasingly important in various industries including, safety, productivity and accuracy.

In the field of artificial intelligence technology, computer vision adds great importance to the AI, without computer vision there is no future to AI.

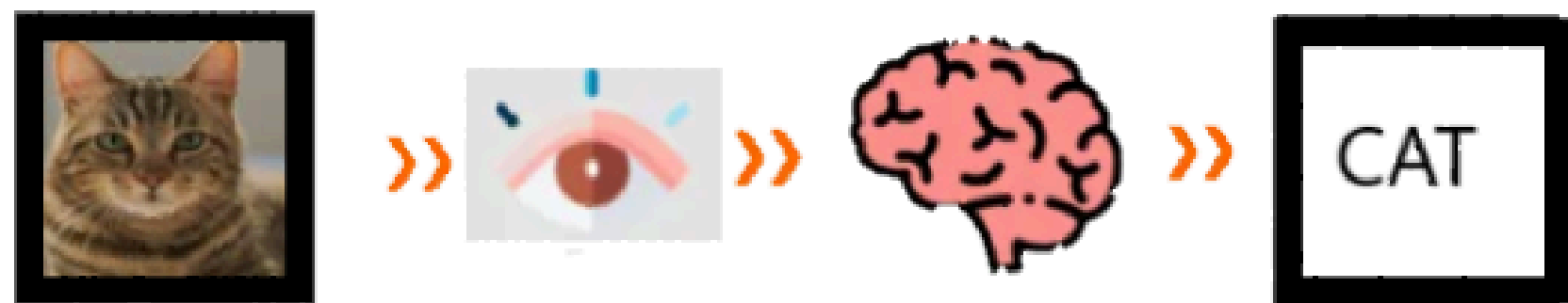
2.5 lakh opportunities.

The Average Salary is nearly 6-8 Lakh per annum.

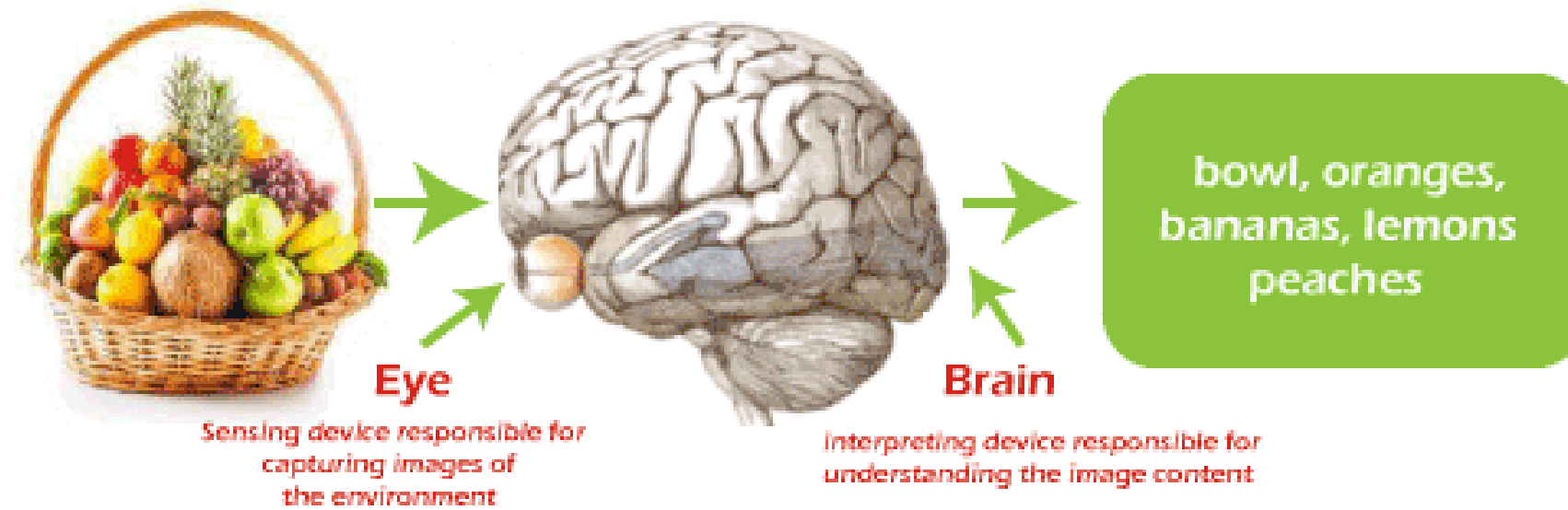
Machine Vision



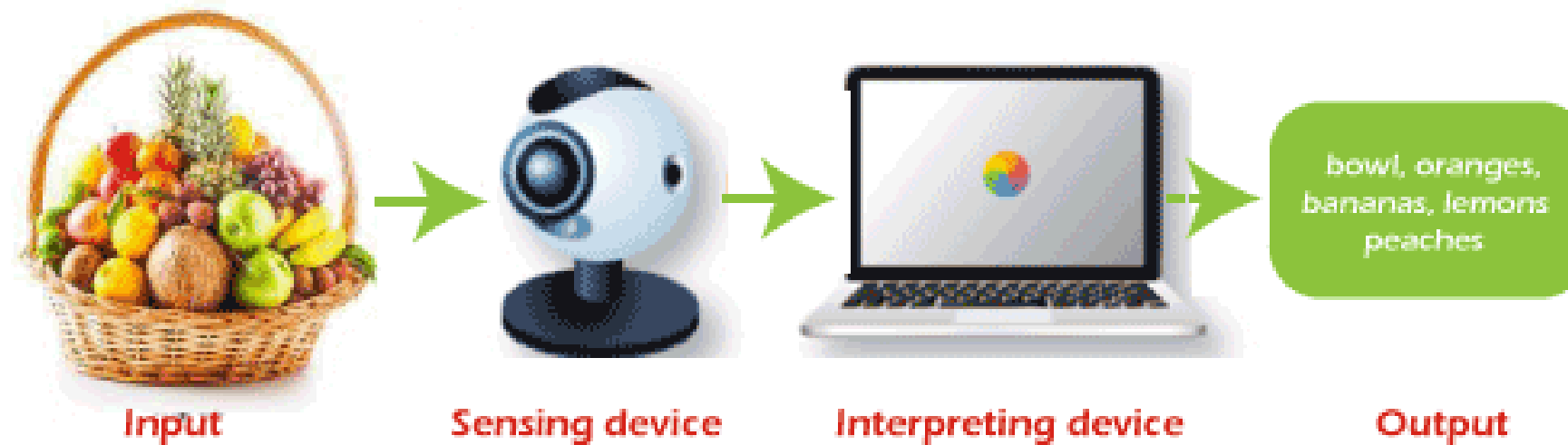
Human Vision



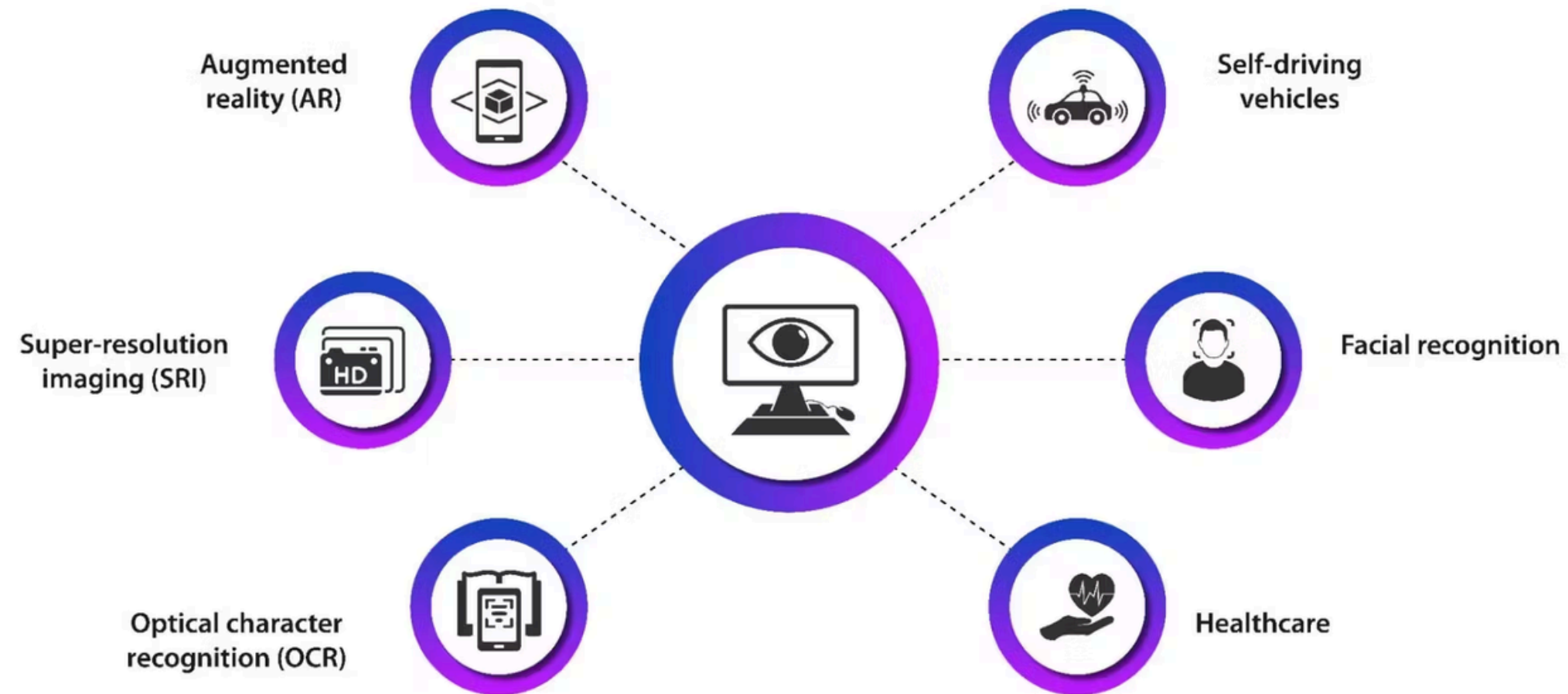
Human Vision System



Computer Vision System



Computer Vision Applications



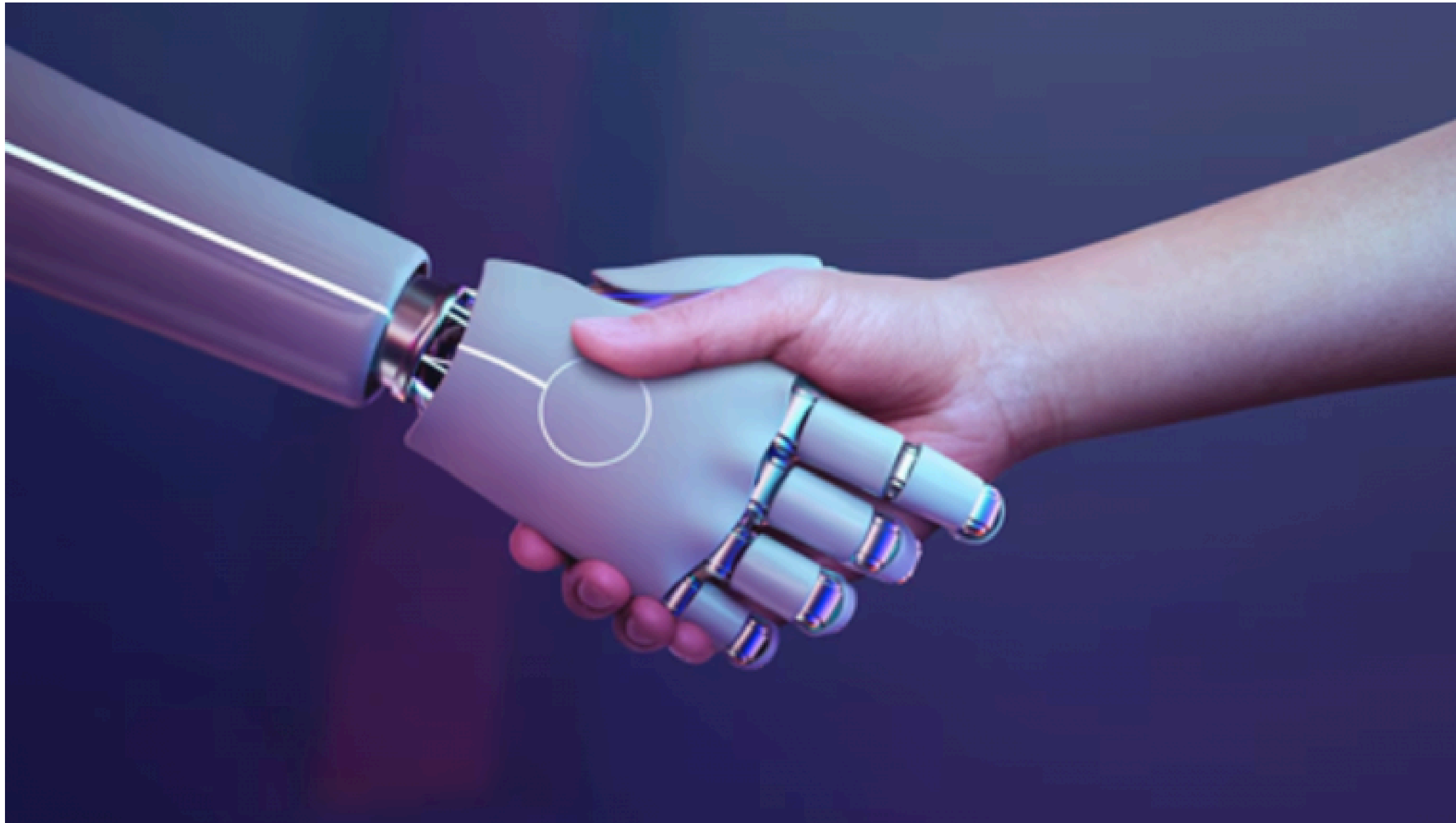
Computer Vision Trends for 2024

Generative AI

Generative AI is becoming a significant trend in computer vision because it allows computers to create realistic images and videos on their own. Instead of relying solely on existing data, generative AI can invent new visual content by learning patterns from the data it's given.

By using generative AI, researchers and developers can make better models, create new visuals, and push the boundaries of what's possible in computer vision. OpenAI's groundbreaking tools, ChatGPT and Dall-E, represent a leap forward in artificial intelligence.

Generative AI



Computer Vision Trends for 2024

Edge computing

Edge computing simplifies the way visual data is processed and analyzed. Unlike traditional methods where data is sent to centralized servers for processing, edge computing brings this processing closer to where the data is generated—right at the “edge” of the network, such as on devices like smartphones, cameras, or sensors.

Edge computing in computer vision enables applications like smart cameras, autonomous vehicles, and IoT devices to make quick, intelligent decisions without always needing a stable internet connection or relying on cloud infrastructure.

Edge computing



Computer Vision Trends for 2024

Merged reality

Merged reality, as a computer vision trend, represents a fusion of physical and virtual worlds through advanced visual processing techniques. In simpler terms, it's about blending real-world environments with digital elements in a seamless manner. This trend leverages computer vision technologies to detect and understand real-world surroundings accurately. It allows virtual objects to interact with them convincingly.

For example, in augmented reality applications, merged reality enables virtual objects to align with real-world surfaces and respond realistically to changes in the environment.

Merged reality



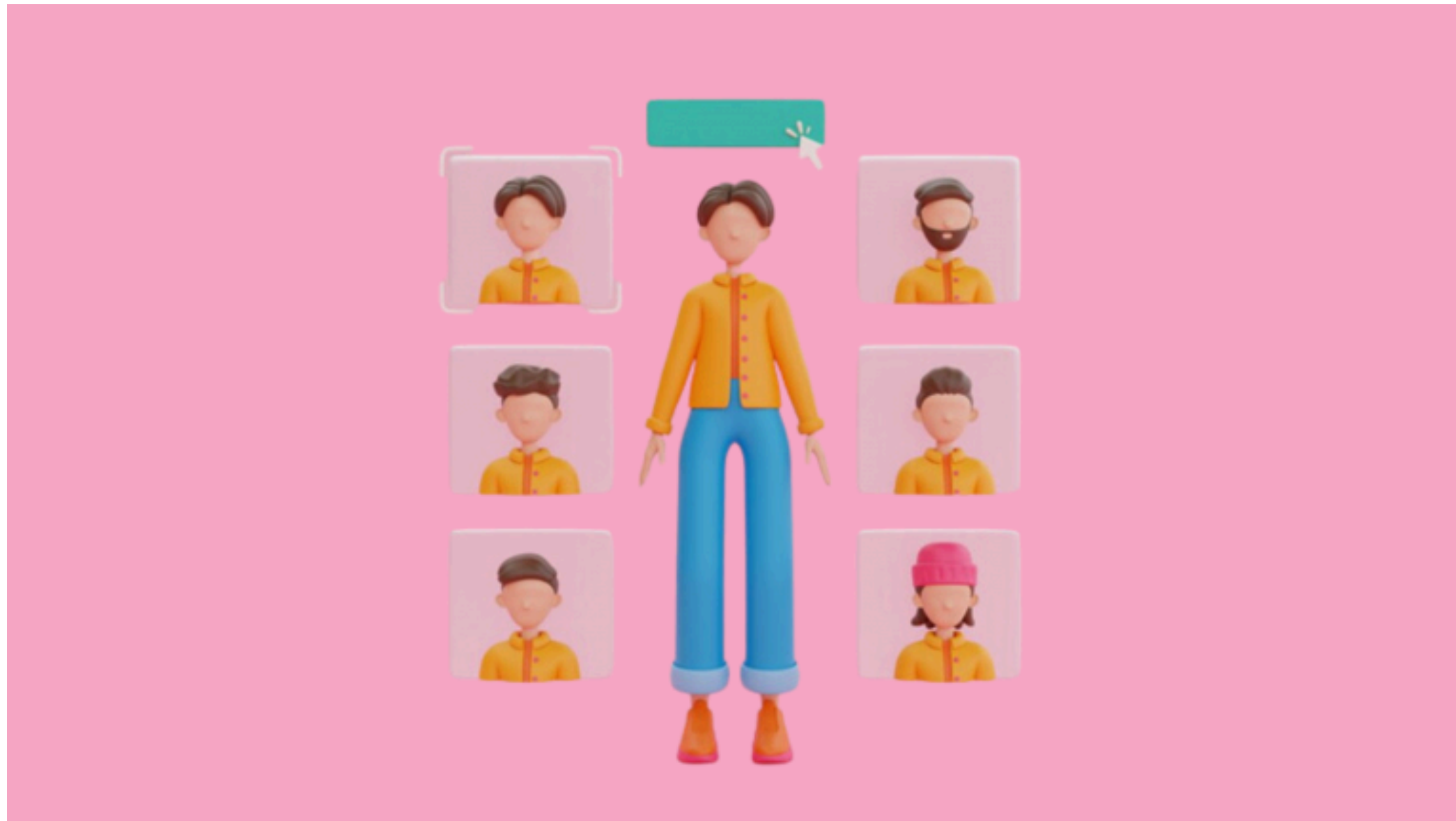
Computer Vision Trends for 2024

3D Modeling

3D modeling is becoming a prominent trend in computer vision because it allows computers to understand and interact with three-dimensional objects in a more lifelike way. Like how we perceive objects in the real world, computer vision systems equipped with 3D modeling can analyze and manipulate objects from multiple angles. This enables more accurate recognition, tracking, and interaction.

This technology is particularly valuable in fields like augmented reality, virtual reality, robotics, and gaming, where realistic representation of 3D objects is essential for immersive user experiences and precise decision-making.

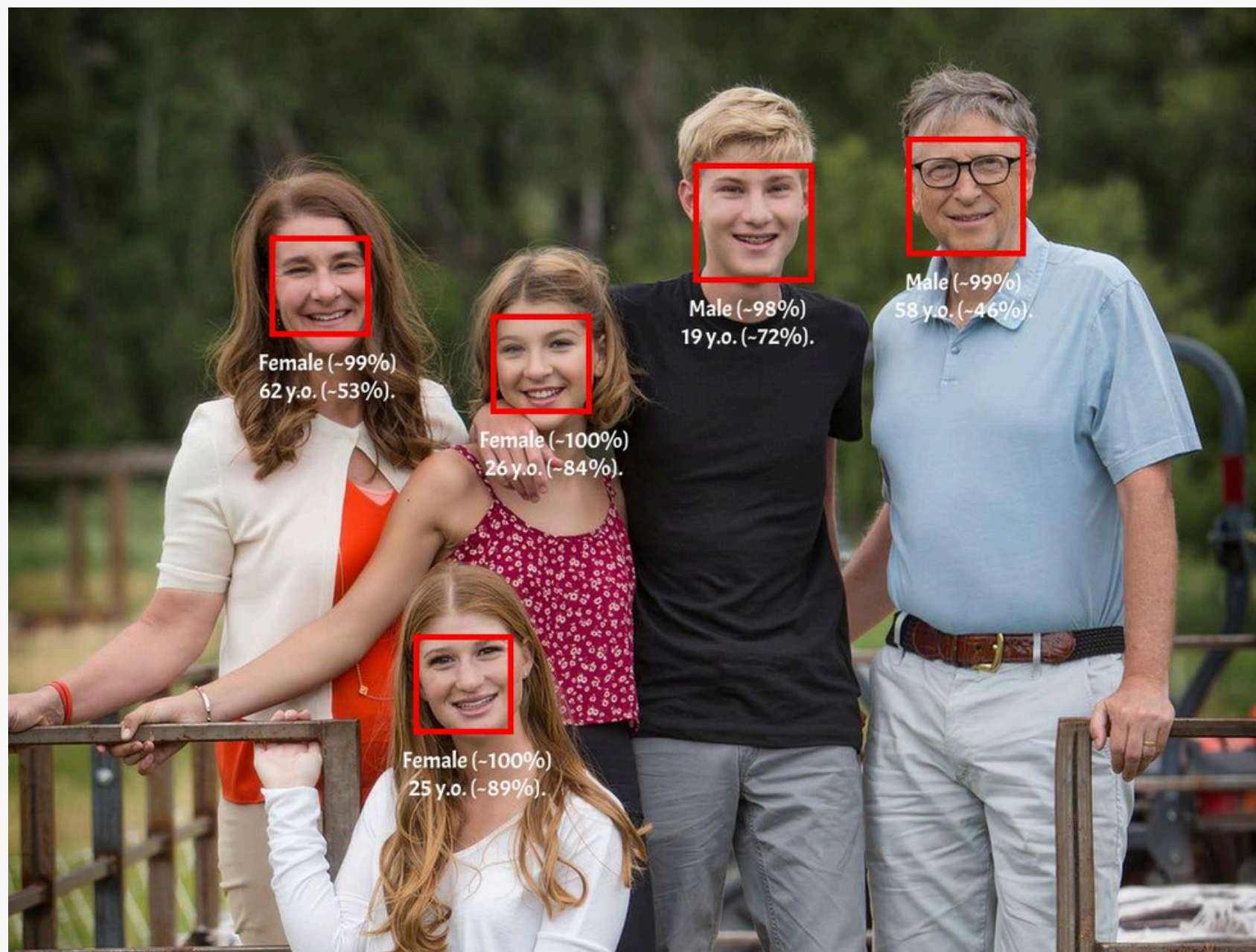
3D Modeling



Face Detection



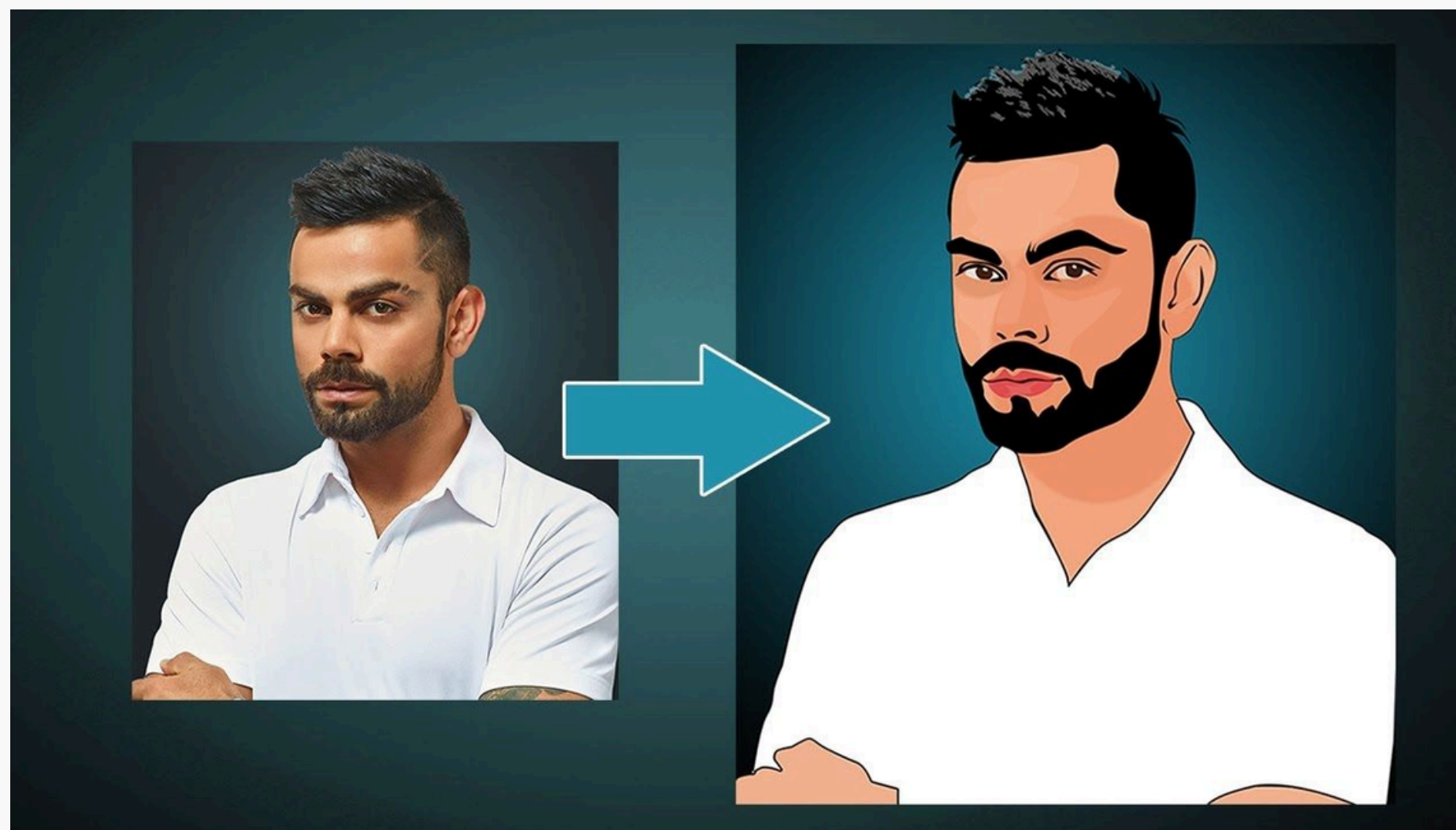
Age & Gender Detection



Face Emotion Detection



Cartoonify image



Installation process of open-cv

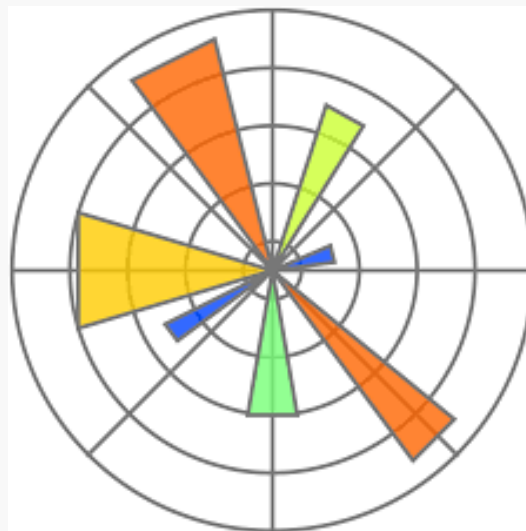
**Python
Software
3.10.4**

**Python
open-cv
package**

**pip install
python
open-cv**

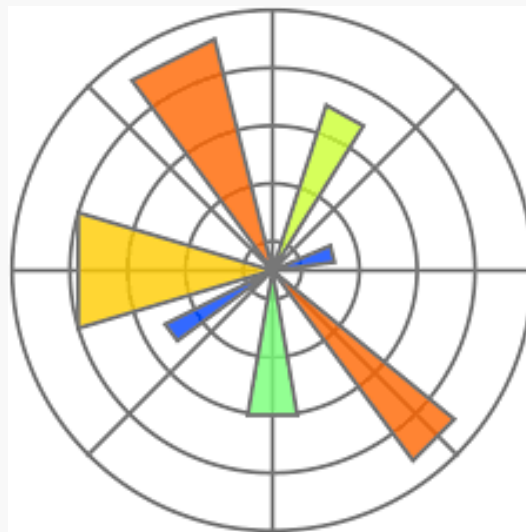
NOW IS
YOUR TIME







Numpy



Numpy





THANK YOU

